

BOLOGNA, Italy

WMO: 161400

Lat: 44.5275N

Lon: 11.2886E

Elev: 38

StdP: 100.88

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.9	(c) -2.2	(d) -9.9	(e) 1.6	(f) 0.5	(g) -8.0	(h) 1.9	(i) 1.1	(j) 7.9	(k) 6.8	(l) 6.3	(m) 5.8	(n) 2.3	(o) 210	(p) 0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	34.9	22.6	33.2	22.2	31.9	21.9	24.5	31.3	23.5	30.4	22.7	29.6	2.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	16.9	28.2	21.2	15.9	27.2	20.2	15.0	26.4	74.0	31.5	70.1	30.6	67.1	29.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.1	(b) 6.1	(c) 5.3	(e) -6.5	(f) 36.9	(g) 2.2	(h) 1.6	(i) -8.1	(j) 38.0	(k) -9.4	(l) 38.9	(m) -10.6	(n) 39.8	(o) -12.3	(p) 40.9		
(a) 7.1	(b) 6.1	(c) 5.3	(e) -7.3	(f) 25.5	(g) 2.1	(h) 1.0	(i) -8.8	(j) 26.2	(k) -10.0	(l) 26.8	(m) -11.2	(n) 27.4	(o) -12.8	(p) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	3.7	5.5	9.9	13.9	18.5	23.0	25.6	25.2	20.3	15.3	9.4	4.4
	DBStd	8.30	2.76	3.33	3.57	3.13	2.98	3.22	2.52	2.86	3.06	3.40	3.56	3.04
	HDD10.0	603	196	127	45	6	0	0	0	0	0	3	51	175
	HDD18.3	2067	453	359	262	136	34	4	0	0	14	106	267	432
	CDD10.0	2289	1	2	42	122	264	391	483	472	310	167	34	1
	CDD18.3	711	0	0	0	3	40	145	224	214	74	11	0	0
	CDH23.3	6896	0	0	3	27	313	1377	2389	2210	541	37	0	0
	CDH26.7	2812	0	0	0	2	63	521	1081	998	144	3	0	0
Wind	WSAvg	2.3	2.0	2.2	2.5	2.6	2.7	2.6	2.7	2.5	2.4	2.1	2.1	1.9
Precipitation	PrecAvg	725	44	45	61	70	64	55	47	57	61	71	79	63
	PrecMax	1067	148	145	172	166	138	119	139	122	193	281	166	131
	PrecMin	420	1	0	1	9	0	13	2	3	1	2	2	9
	PrecStd	161	36	35	34	45	38	30	37	37	47	60	44	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.9	22.8	26.2	30.8	34.9	36.7	37.2	32.1	26.5	20.2	14.1
		MCWB	9.1	9.8	13.8	17.1	20.2	22.2	21.9	22.2	21.4	18.8	15.2	10.3
	2%	DB	11.1	14.2	20.0	23.8	28.2	33.1	34.8	35.1	30.1	24.1	17.8	11.5
		MCWB	7.4	9.0	12.5	15.5	19.0	22.2	22.6	22.7	20.8	18.0	14.2	8.9
	5%	DB	9.3	12.8	18.2	22.0	26.8	31.2	33.2	33.2	28.2	22.4	15.9	10.0
		MCWB	6.6	8.4	11.7	14.6	18.4	21.5	22.2	22.4	19.9	17.1	13.2	8.1
	10%	DB	7.9	11.0	16.3	20.1	25.0	29.8	31.9	31.8	26.6	20.8	14.2	8.9
		MCWB	6.0	7.4	10.8	13.7	17.5	20.8	21.8	21.9	18.8	16.2	12.2	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	11.2	14.4	17.7	21.5	24.4	25.6	26.0	22.9	20.3	16.3	11.6
		MCDB	12.4	15.2	20.2	24.6	27.9	31.4	33.3	33.2	28.8	23.6	18.0	13.5
	2%	WB	8.3	9.9	13.2	16.3	20.1	23.1	24.3	24.7	21.9	19.0	14.9	9.7
		MCDB	9.9	13.1	18.4	22.0	26.5	30.5	31.4	31.3	27.4	22.3	17.1	10.5
	5%	WB	7.1	8.9	12.3	15.2	19.1	22.3	23.5	23.6	20.9	17.8	13.8	8.4
		MCDB	8.4	11.7	17.1	20.6	25.0	29.5	30.5	30.4	26.5	21.4	15.6	9.5
	10%	WB	6.0	7.9	11.3	14.3	18.1	21.5	22.6	22.7	19.9	16.9	12.6	7.2
		MCDB	7.4	10.1	15.5	19.0	23.7	28.3	29.6	29.4	25.1	20.2	14.3	8.3
Mean Daily Temperature Range	5% DB	MDBR	6.4	8.1	9.8	9.8	10.2	10.7	11.2	11.3	10.1	8.0	6.2	6.1
		MCDBR	9.4	11.0	12.8	12.5	12.3	12.5	12.6	12.9	11.9	9.9	8.2	8.0
		MCWBR	6.6	7.1	7.3	6.6	6.0	5.5	5.4	5.3	5.2	5.2	5.3	5.7
	5% WB	MCDBR	7.6	9.7	11.3	11.1	11.0	11.5	11.3	11.4	10.4	8.6	6.8	6.3
		MCWBR	5.9	6.6	6.9	6.2	5.9	5.6	5.5	5.4	5.1	4.8	4.7	4.9
Clear-Sky Solar Irradiance	taub	0.369	0.396	0.443	0.465	0.463	0.478	0.466	0.455	0.442	0.442	0.399	0.363	
	taud	2.308	2.198	2.078	2.057	2.109	2.118	2.163	2.211	2.214	2.167	2.227	2.344	
	Ebn at Noon	723	767	779	798	814	801	806	800	774	705	671	690	
	Edh at Noon	84	111	142	158	154	153	145	134	123	114	91	75	
All-Sky Solar Radiation	RadAvg	1.38	2.23	3.51	4.60	5.74	6.54	6.74	5.77	4.22	2.57	1.48	1.21	
	RadStd	0.21	0.33	0.49	0.46	0.50	0.38	0.36	0.37	0.30	0.33	0.20	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.97	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+67	N/A

Nomenclature: See separate page